Process–Voltage–Temperature Analysis of a 6T SRAM Bitcell and a 128×128 Array Macro

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Abstract

We characterize a 6T SRAM bitcell across three process–voltage corners (FF 1.2 V, TT 1.0 V, SS 1.0 V, with SS 0.8 V where the metric is defined) and bridge the cell-level evidence to a 128×128, 8-bit, 16:1 column-multiplexed macro modeled in NVSim. Stability and writability are extracted from butterfly curves (largest-square limiting eye for read and hold SNM, diagonal-corner write noise margin) and from a transient pulse-window read-disturb metric. We compare the baseline cell against three optimizations: a high-$\boldsymbol{V_t}$ access-transistor variant, a negative-bitline write assist, and a wordline-underdrive read assist. The optimized cells trade margin for power or speed in expected ways: high-$\boldsymbol{V_t}$ reduces TT hold-window energy by 13.9 % at the cost of a 10.2 % read-SNM penalty; negative-bitline cuts TT write delay by 54.5 %; and wordline-underdrive raises TT read SNM by 39.9 % while suppressing read disturb by 57.3 %. Every macro number is labeled by its evidence type (*Estimated by NVSim*, *Measured in Cadence*, or *Derived proxy*) so that NVSim outputs are never silently mixed with Cadence-derived bridges.

Keywords: SRAM, 6T bitcell, PVT, SNM, write noise margin, NVSim, high-$\boldsymbol{V_t}$, negative bitline, wordline underdrive

1 Introduction

Static random-access memory (SRAM) sets the area, energy, and access-time floor of most modern systems-on-chip, and its margins are the part of the chip most exposed to process–voltage–temperature (PVT) variation. A six-transistor (6T) bitcell that holds reliably under one corner can lose read or write margin at another, and assist circuits added to recover that margin almost always trade off against either dynamic energy, leakage, or speed.

This work quantifies those trade-offs. We start from a single 6T bitcell, characterize its hold static noise margin (SNM), read SNM, and write noise margin (WNM) across the corners that matter, and then evaluate three commonly proposed optimizations against the same baseline:

1. **High-V_t** access transistors, intended to reduce hold-window leakage at the cost of read-SNM.
2. **Negative-bitline** write assist, intended to accelerate the write transition.
3. **Wordline-underdrive** read assist, intended to recover read SNM and suppress read disturb.

The cell-level evidence is then bridged into a 128×128, 8-bit, 16:1 column-multiplexed macro modeled in NVSim. Because NVSim does not

natively model the three optimizations, we keep every macro-level number labeled with its evidence type — *Estimated by NVSim*, *Measured in Cadence*, or *Derived proxy* — so that simulator outputs are never silently mixed with Cadence-derived bridges.

2 Methodology

2.1 Bitcell SNM and WNM extraction

Hold SNM and read SNM are extracted from the standard butterfly DC sweeps of the storage node V_Q versus the complementary node $V_{\overline{Q}}$. We use the largest-square limiting-eye fit: the SNM is the side length of the largest square that fits inside the smaller of the two eyes formed by the inverter and access-transistor characteristics. This is the conservative choice when the two lobes are asymmetric and is the convention used throughout the figures in Section 3.

Write noise margin uses the diagonal-corner $V_1 = V_2$ convention: WNM is the side length of the largest square anchored on the diagonal that fits between the read and write curves of the cell during a write. The same convention is used uniformly across the WNM butterfly figures, so the reported numbers are directly comparable across corners and across optimizations.

All bitcell metrics in this report are labeled *Measured in Cadence*.

2.2 Wordline-underdrive read disturb

The wordline-underdrive read-disturb value is a transient pulse-window metric, not a DC quantity. It is the maximum disturbance of the storage node opposite to the access path, measured on $\overline{Q}$ during the read pulse window of the transient bench. This metric captures the dynamic node bounce that is the actual failure mechanism a wordline-underdrive assist is designed to suppress, and it is reported alongside the static read-SNM gain to avoid claiming a benefit that exists only in DC.

2.3 Cadence-to-NVSim bridge

The macro evaluation in Sections 5 and 6 is anchored on a single direct NVSim run for the 128×128, 8-bit, 16:1 baseline organization. From that anchor we apply optimization-specific changes to exactly the columns where the Cadence evidence justifies a change, and leave the rest at the NVSim baseline. Every row of the bridge carries one of three labels:

- *Estimated by NVSim* — copied directly from the NVSim baseline run.
- *Measured in Cadence* — comes from the bitcell simulations and is included only as supporting evidence, not as a macro number.
- *Derived proxy* — a baseline NVSim number scaled by a Cadence-measured ratio at TT nominal, used only when there is no NVSim knob for the optimization in question.

The TT-nominal corner is the bridge anchor; corner robustness is left to the bitcell section. Concretely, only two macro columns receive a derived proxy in this report: the High-V_t leakage power (scaled by the TT hold-window energy ratio) and the Negative-BL write latency (scaled by the TT write-delay ratio). Wordline-underdrive does not change any macro column because the assist's defended benefit is cell-level stability, not a macro timing or energy shift; we therefore present it in the bitcell results section only.

3 Baseline bitcell stability and writability

Hold SNM is comfortable at all three voltage–corner combinations (422–461 mV), confirming that the cell holds robustly without assist. Read SNM is the smallest of the three margins (212–224 mV) and sets the read-side reliability floor. WNM at TT nominal is 362 mV; the SS 1.0 V corner is the worst case at 259 mV, which motivates the negative-bitline write assist evaluated in Section 4.2. Per-corner butterfly overlays for hold SNM, read SNM, and WNM are reproduced in Appendix A.

4 Optimization studies

4.1 High-V_t access transistors

Raising the access-transistor threshold voltage cuts subthreshold leakage during the hold window, but it also weakens the access path and reduces read-SNM because the storage node is pulled less

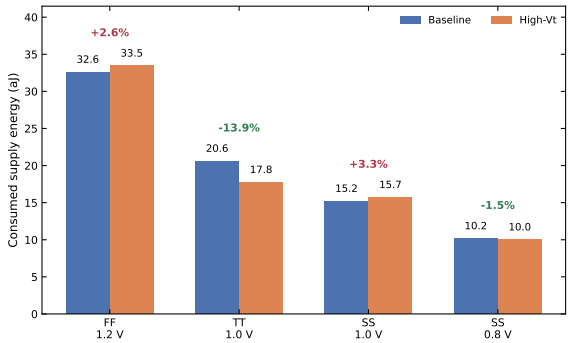

Fig. 1 Per-corner hold-window supply energy (6–9 ns integration window) for the baseline cell versus the high-V_t variant. TT 1.0 V improves by 13.9 %; FF 1.2 V regresses slightly because at the FF corner the leakage path is dominated by gate leakage that is not improved by raising V_t.

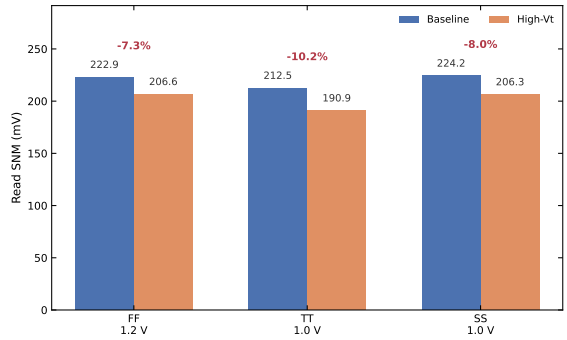

Fig. 2 Per-corner read SNM for the baseline cell versus the high-V_t variant, styled to pair visually with Figure 1. SS 0.8 V is intentionally absent because the baseline cell does not have a valid read butterfly at that supply without a wordline-underdrive assist; the limiting-eye SNM is therefore undefined.

stiffly during a read. Figures 1 and 2 report the two sides of this trade-off across corners.

The TT-corner numbers summarize the trade-off cleanly: hold-window supply energy drops from 20.6 aJ to 17.8 aJ (-13.9 %), while read SNM drops from 212 mV to 191 mV (-10.2 %). Hold SNM degrades by less than 4 % at TT and by less than 3 % at SS 1.0 V. The optimization is consistent with its intent at the typical and slow corners; the FF corner energy regression flags that high-V_t is not a universal leakage win and motivates corner-aware deployment. Per-corner hold-SNM and read-SNM butterfly overlays are reproduced in Appendix A.

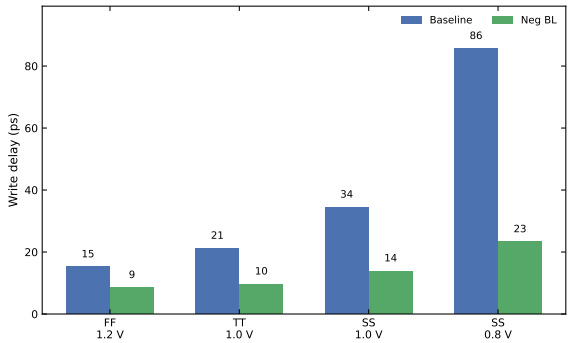

Fig. 3 Per-corner WL-to-Q write delay for the baseline cell versus the negative-bitline assist. TT improves by 54.5 %; SS 0.8 V improves by 72.8 %, the largest gain because that corner is the slowest baseline.

4.2 Negative-bitline write assist

Driving the bitline below ground during the write strengthens the access transistor's pull-down ability and accelerates the storage-node flip. Figure 3 reports the WL-to-Q write delay for the assisted cell versus the baseline across all four corners.

The benefit grows monotonically as the corner slows, which is exactly the corner profile a write assist is supposed to have. The write-delay improvement at SS 0.8 V (-72.8 %, from 85.6 ps to 23.3 ps) is the largest single-knob improvement reported in this work. The full per-corner waveform grid is reproduced in Appendix B.

4.3 Wordline-underdrive read assist

Underdriving the wordline during a read keeps the access transistors weaker than the cross-coupled pull-down devices, which raises read SNM and suppresses the dynamic disturbance of the storage node. Figures 4 and 5 report the static and transient sides of this assist.

The static and transient evidence agree, which is the relevant cross-check: a read-SNM gain that did not show up in the transient read-disturb window would not be a real benefit. Both metrics improve at every corner, with the largest gains at the slowest, lowest-supply corner (SS 0.8 V). The all-corner read-SNM butterfly overlay is reproduced in Appendix A.

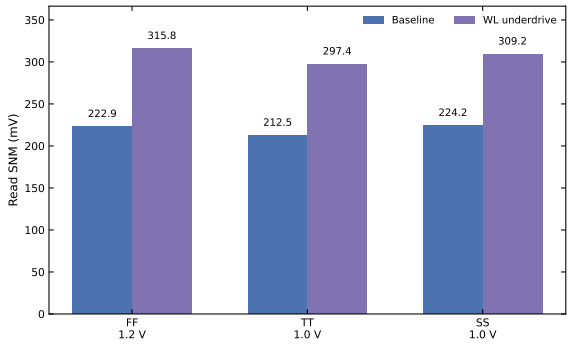

Fig. 4 Per-corner read SNM for the baseline cell versus the wordline-underdrive assist. TT improves by 39.9 % (212 mV to 297 mV); FF and SS 1.0 V improve by similar margins.

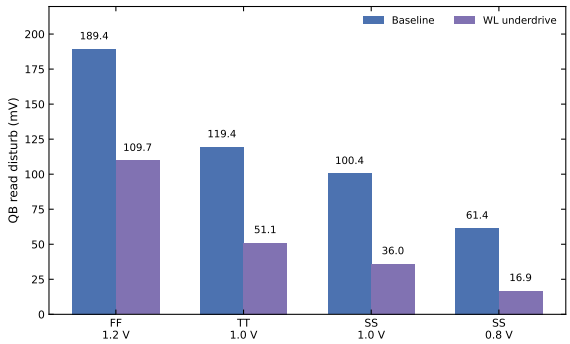

Fig. 5 Per-corner pulse-window transient read-disturb metric (Section 2.2) for the baseline cell versus the wordline-underdrive assist. TT read disturb drops by 57.3 %; SS 0.8 V drops by 72.5 %.

5 Baseline NVSim macro

Table 1 reports the direct NVSim baseline for the 128×128, 8-bit, 16:1 organization. The full read-access latency budget breaks down as in Figure 6.

The breakdown makes clear that the row-decode path is where most of the macro's latency lives, and it is unaffected by any of the three bit-cell optimizations evaluated in this report. That observation constrains which macro columns can plausibly move under each optimization.

6 Bridged optimization results

Per the bridge methodology of Section 2.3, only two macro columns move under the optimizations evaluated here: the high-V_t leakage power and the negative-bitline write latency. Both are derived proxies anchored on the TT Cadence ratios; all other macro columns are reused from the NVSim baseline.

Table 1 Baseline NVSim macro summary at the locked organization (128×128, 8-bit, 16:1). All rows are *Estimated by NVSim*.

Baseline macro quantity	Value
Capacity	2048 B
Physical array	128 × 128
Word width	8 bits
Column selection	16:1
Area	0.006197 mm^2
Read latency	0.8821 ns
Write latency	0.8821 ns
Read energy	1.3540 pJ
Write energy	0.3210 pJ
Leakage	0.4854 μW

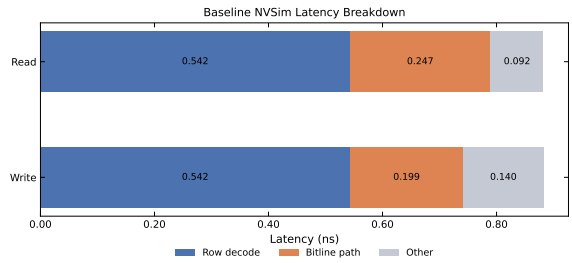

Fig. 6 Latency budget for the baseline macro. Row decode (predecoder + row decoder) dominates both reads and writes at ∼61 % of total latency; the bitline path accounts for ∼28 % of read latency and ∼23 % of write latency.

6.1 High-V_t macro leakage

6.2 Negative-BL macro write latency

6.3 Why wordline-underdrive does not move a macro column

The wordline-underdrive assist is real and well-supported by the cell-level evidence in Section 4.3, but its benefit is read-SNM and read-disturb suppression, not a macro timing or energy shift that the bridge can defend. Forcing a derived-proxy column onto NVSim's macro row would therefore overclaim. The assist is presented in the bitcell results section and intentionally omitted from this section.

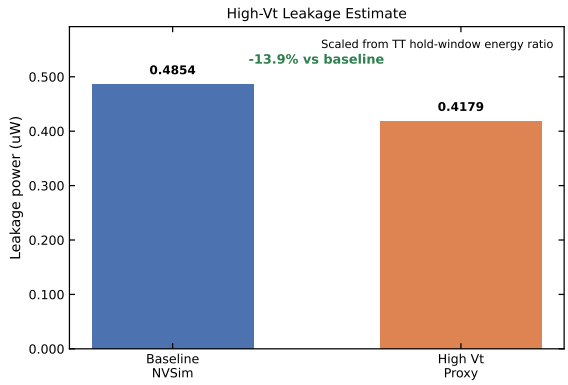

Fig. 7 High-V_t macro leakage power as a derived proxy: the baseline NVSim leakage (0.4854 μW) scaled by the TT hold-window energy ratio (0.8610) gives 0.4179 μW (-13.9 %). Area, latency, and dynamic energy stay at the NVSim baseline.

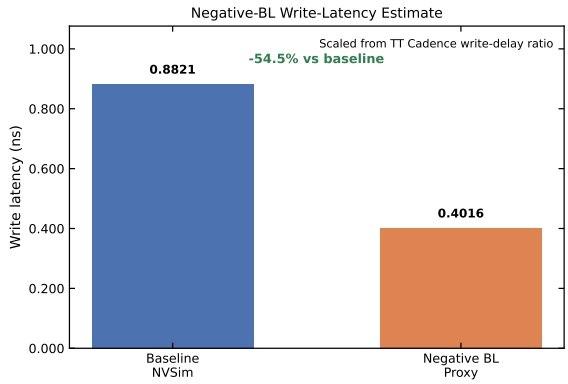

Fig. 8 Negative-BL macro write latency as a derived proxy: the baseline NVSim write latency (0.8821 ns) scaled by the TT Cadence write-delay ratio (0.4553) gives 0.4016 ns (-54.5 %). Read latency, energy, and leakage stay at the NVSim baseline.

7 Cadence verification

Cadence verification of the 128×128 macro at the schematic level is shown in Figures 9–10. The screenshots confirm that the baseline organization assumed by NVSim (128 rows, 128 physical columns, 8-bit words, 16:1 column multiplexing) is the same organization reflected in the schematic and that the baseline access waveforms behave as intended.

Fig. 9 Cadence schematic of the 128×128 array (placeholder for screenshot).

Fig. 10 Cadence baseline access waveforms on the 128×128 macro (placeholder for screenshot).

8 Limitations and evidence boundaries

A few boundaries on the evidence in this report are worth stating explicitly:

- TT nominal is the bridge anchor. Corner robustness lives in the bitcell section (Sections 3 and 4); we do not attempt to scale the macro across corners because the NVSim baseline is itself a single-corner run.

- Negative-BL and wordline-underdrive were not directly simulated as full-array NVSim cases. Their cell-level benefits are real, but the macro-level claims in Section 6 are bounded to what a TT-anchored derived proxy can defend.
- Archived NVSim material (under `old_nvsim/` in the project tree) is not part of the live evidence used for this report. The single live anchor is the fresh baseline run cited in Table 1.
- Cadence verification (Section 7) is a sanity check on the baseline organization, not an independent measurement of the macro-level optimization deltas.

9 Conclusion

Across three corners (and four where the metric is defined), the baseline 6T cell holds robustly, but its read SNM is the binding margin and its SS 1.0 V WNM is the binding writability margin. High-V_t buys a 13.9 % TT hold-window energy reduction at a 10.2 % read-SNM cost; negative-bitline cuts TT write delay by 54.5 %; wordline-underdrive raises TT read SNM by 39.9 % and suppresses transient read disturb by 57.3 %. Bridged into a 128×128 macro, only the high-V_t leakage and negative-BL write latency move; everything else stays pinned to the NVSim baseline. The evidence-type labeling makes those boundaries explicit, which is the contribution we expect to be most useful when the same flow is reused on a different cell or a different macro.

Author contributions

Author contributions are pending and will be provided in a future revision.

Declarations

- **Funding:** Not applicable. This work was carried out as a coursework project for ESE 5760 at the University of Pennsylvania.
- **Conflict of interest:** The authors declare no conflict of interest.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** All extracted metrics and bridge tables are available in the project repository (`sram_analysis_plots/`).
- **Materials availability:** Not applicable.
- **Code availability:** Plot and analysis scripts are available in the project repository (`scripts/`).

Appendix A Bitcell butterfly and overlay figures

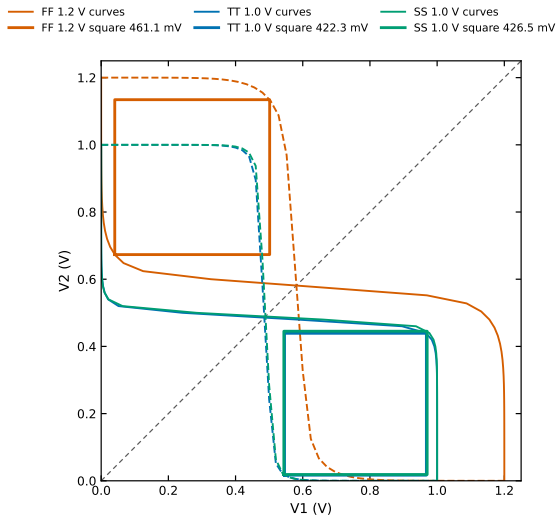

Fig. A.1 Baseline hold-SNM butterfly overlays across corners.

Appendix B Negative-BL waveform grid

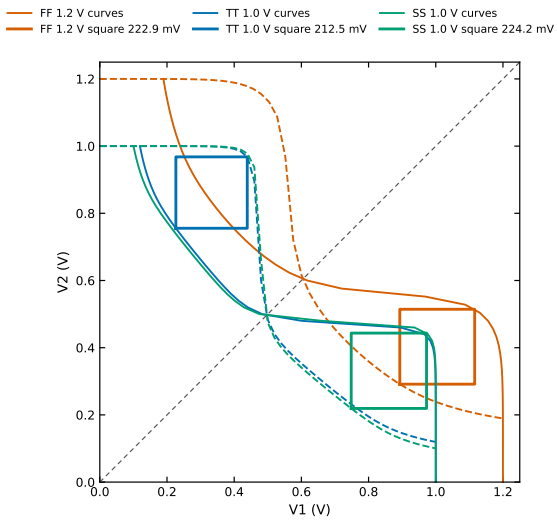

Fig. A.2 Baseline read-SNM butterfly overlays across corners.

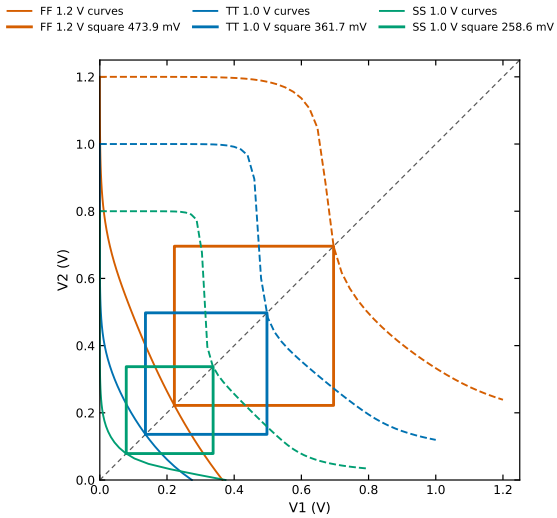

Fig. A.3 Baseline WNM butterfly overlays across corners.

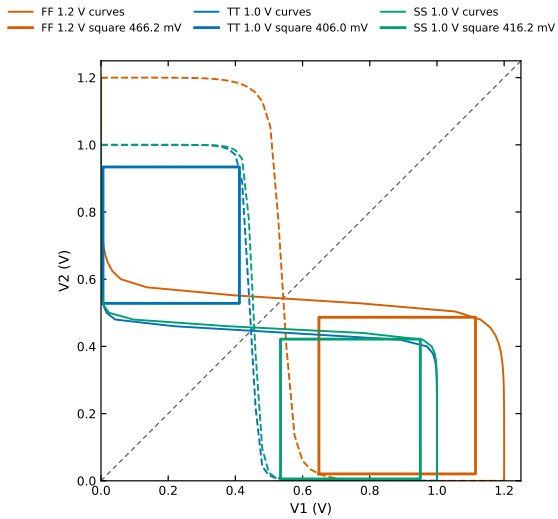

Fig. A.4 High-V_t hold-SNM butterfly overlays across corners.

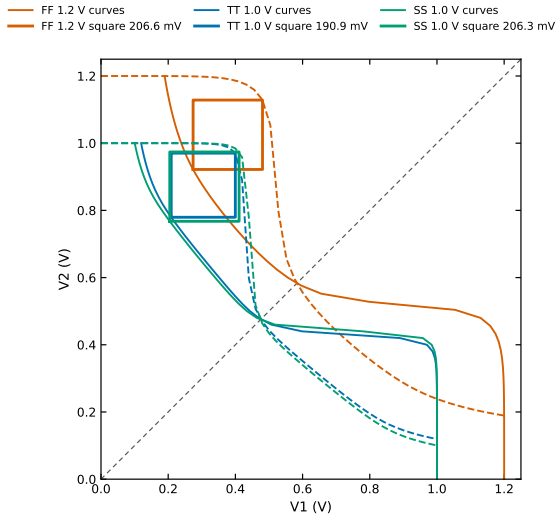

Fig. A.5 High-V_t read-SNM butterfly overlays across corners.

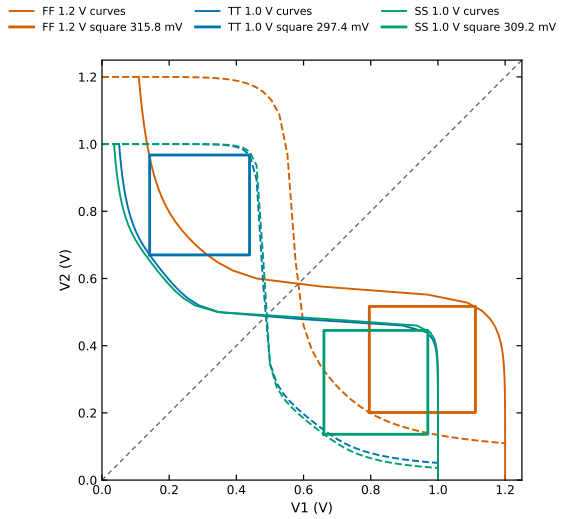

Fig. A.6 Wordline-underdrive read-SNM butterfly overlays across corners.

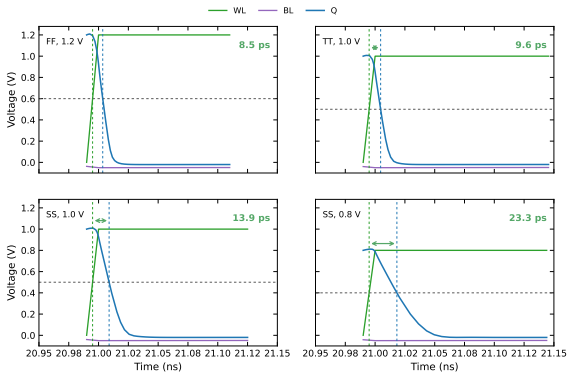

Fig. B.1 Per-corner negative-bitline write-delay waveform grid.